AP Calculus BC — Unit 5 Review Book

Analytical Applications of Differentiation

This review book is designed for a student who wants a full, beginner-friendly final review of Unit 5. It is detailed, but organized for fast exam preparation. Every major AP skill is included: theorem conditions, critical points, increasing/decreasing, local and absolute extrema, concavity, graph relationships, optimization, and implicit-relation behavior.

How to use this book: First learn the idea and the key formulas. Then study the worked example in each section. Finally, memorize the common mistakes so you can avoid the most frequent AP errors.

Unit 5 Master Checklist

- Understand and apply the Mean Value Theorem and Rolle's Theorem.
- Find and classify critical numbers.
- Determine intervals of increase/decrease and concavity.
- Use the First and Second Derivative Tests correctly.
- Find absolute extrema on closed intervals with the Candidates Test.
- Translate word problems into optimization functions and constraints.
- Analyze implicit relations and identify horizontal and vertical tangents.

5.1 Using the Mean Value Theorem

Concept

The **Mean Value Theorem (MVT)** connects an average rate of change on an interval to an instantaneous rate of change at some point inside the interval.

$$f'(c) = \frac{f(b) - f(a)}{b - a} \quad \text{for some } c \in (a, b)$$

The theorem applies only when the function is **continuous on** [a,b] and **differentiable on** (a,b).

$$\text{continuous on } [a, b], \qquad \text{differentiable on } (a, b)$$

The geometric meaning is that some tangent line is parallel to the secant line through (a,f(a)) and (b,f(b)).

What AP asks

Check whether the hypotheses of the theorem are satisfied.

Find all values of *c* that satisfy the theorem.

Explain the theorem in words in a real or graphical context.

Rolle's Theorem

Rolle's Theorem is a special case of the MVT. If f(a)=f(b), then the secant slope is zero, so there must be a point where the tangent slope is zero.

$$f(a) = f(b) \Rightarrow f'(c) = 0 \text{ for some } c \in (a, b)$$

Worked Example

Find all values of *c* guaranteed by the Mean Value Theorem for f(x)=x^2-4x+1 on [1,3].

$$\frac{f(3) - f(1)}{3 - 1} = \frac{(-2) - (-2)}{2} = 0$$

$$f'(x) = 2x - 4$$

$$2c - 4 = 0 \Rightarrow c = 2$$

So the required value is **c = 2**.

Common mistakes

Forgetting to check continuity and differentiability first.

Including endpoints when solving for *c*. The theorem requires *c* to be inside the open interval.

Thinking the theorem tells you where every parallel tangent occurs. It only guarantees at least one.

5.2 Extreme Value Theorem, Critical Points, and Local vs. Absolute Extrema

Concept

The **Extreme Value Theorem (EVT)** says that a function continuous on a closed interval must attain both an absolute maximum and an absolute minimum on that interval.

$$\text{continuous on } [a,b] \Rightarrow \text{absolute max and absolute min exist on } [a,b]$$

A **critical number** is a number c in the domain of f such that either f'(c)=0 or f'(c) does not exist.

$$f'(c) = 0 \quad \text{or} \quad f'(c) \text{ DNE}$$

Key vocabulary

Local maximum/minimum: largest or smallest value in a small neighborhood.

Absolute maximum/minimum: largest or smallest value on the entire interval or domain.

Worked Example

Find the critical numbers of f(x)=x^3-3x.

$$f'(x) = 3x^2 - 3 = 3(x^2 - 1)$$
$$3(x^2 - 1) = 0 \Rightarrow x = \pm 1$$

The critical numbers are **-1** and **1**.

Common mistakes

Listing a point that is not in the domain as a critical number.

Assuming every critical number is automatically a local extremum.

Confusing the existence of extrema with a method for finding them.

5.3 Determining Intervals on Which a Function Is Increasing or Decreasing

Concept

The sign of the first derivative determines whether a function increases or decreases.

$$f'(x) > 0 \Rightarrow f \text{ is increasing}$$
$$f'(x) < 0 \Rightarrow f \text{ is decreasing}$$

Procedure

1. Compute f'(x).
2. Find critical numbers and points where f'(x) is undefined.
3. Split the number line into intervals.
4. Test the sign of f'(x) on each interval.

Worked Example

For f'(x)=(x-1)(x+2), determine where f is increasing.

Critical numbers: x = -2 and x = 1.

$$x < -2: \ f'(x) > 0; \qquad -2 < x < 1: \ f'(x) < 0; \qquad x > 1: \ f'(x) > 0$$

So f is increasing on (-\infty,-2)\cup(1,\infty).

Common mistakes

Using the sign of f(x) instead of the sign of f'(x).

Stopping after finding critical numbers without doing a sign chart.

Treating any zero of f'(x) as an extremum without checking sign changes.

5.4 Using the First Derivative Test to Find Relative (Local) Extrema

Concept

The **First Derivative Test** uses sign changes in f'(x) around a critical point.

$$f': \ + \rightarrow - \ \Rightarrow \text{local maximum}$$
$$f': \ - \rightarrow + \ \Rightarrow \text{local minimum}$$
$$f': \ \text{no sign change} \Rightarrow \text{neither}$$

Worked Example

Suppose f'(x) changes from negative to positive at x=3. Then f decreases before x=3 and increases after x=3, so x=3 is a local minimum.

Hard Example

For f(x)=x^4,

$$f'(x) = 4x^3$$

At x=0, f'(0)=0, but the sign of f' changes from negative to positive, so x=0 is a local minimum.

Common mistakes

Declaring a local extremum just because f'(c)=0.

Ignoring the sign behavior on both sides of the critical point.

5.5 Using the Candidates Test to Find Absolute (Global) Extrema

Concept

On a closed interval, absolute extrema must occur at either **endpoints** or **critical numbers** inside the interval. This is called the **Candidates Test**.

Procedure

1. Compute f'(x).
2. Find all critical numbers in the interval.
3. Evaluate f at each critical number and each endpoint.
4. Compare the function values.

Worked Example

Find the absolute extrema of f(x)=x^3-3x on [-2,2].

$$f'(x) = 3x^2 - 3 = 0 \Rightarrow x = \pm 1$$
$$f(-2) = -2, \quad f(-1) = 2, \quad f(1) = -2, \quad f(2) = 2$$

So the absolute maximum value is 2 at x=-1 and x=2, and the absolute minimum value is -2 at x = -2 and x = 1.

Common mistakes

Forgetting endpoints.

Comparing derivative values instead of function values.

Using the Candidates Test on an open interval without extra analysis.

5.6 Determining Concavity of Functions over Their Domains

Concept

The sign of the second derivative determines concavity.

$$f''(x) > 0 \Rightarrow f \text{ is concave up}$$
$$f''(x) < 0 \Rightarrow f \text{ is concave down}$$

Inflection points

An **inflection point** occurs where concavity changes.

Usually you first find candidates by solving f''(x)=0 or locating points where f'' is undefined, then check whether the sign of f'' actually changes.

Worked Example

For f''(x)=x-4,

$$x < 4 \Rightarrow f''(x) < 0 \Rightarrow \text{concave down}$$
$$x > 4 \Rightarrow f''(x) > 0 \Rightarrow \text{concave up}$$

So x=4 is an inflection point.

Common mistakes

Using the sign of f(x) or f'(x) instead of f''(x).

Assuming every solution of f''(x)=0 is automatically an inflection point.

5.7 Using the Second Derivative Test to Find Extrema

Concept

If f'(c)=0, then the sign of f''(c) can classify the critical point.

$$f''(c) > 0 \Rightarrow \text{local minimum}$$

$$f''(c) < 0 \Rightarrow \text{local maximum}$$
$$f''(c) = 0 \Rightarrow \text{inconclusive}$$

Worked Example

For f(x)=x^3-3x^2,

$$f'(x) = 3x^2 - 6x = 3x(x-2)$$

So the critical numbers are x=0 and x=2.

$$f''(x) = 6x - 6$$
$$f''(0) = -6 < 0 \Rightarrow \text{local maximum at } x = 0$$
$$f''(2) = 6 > 0 \Rightarrow \text{local minimum at } x = 2$$

Common mistakes

Using the test when f'(c)\neq 0.

Thinking f''(c)=0 means no extremum. It only means the test fails.

5.8 Sketching Graphs of Functions and Their Derivatives

Concept

You must be able to connect the graph of f with the graph of f'.

If f is increasing, then f' is positive.

If f has a horizontal tangent, then f'=0 there.

If f is concave up, then f' is increasing.

Useful correspondences

$$f \text{ increasing} \Longleftrightarrow f' > 0$$
$$f \text{ decreasing} \Longleftrightarrow f' < 0$$

$$f \text{ concave up} \Longleftrightarrow f' \text{ increasing}$$
$$f \text{ concave down} \Longleftrightarrow f' \text{ decreasing}$$

Worked Example

If f has a local maximum at x=a, then f' changes from positive to negative at x=a.

If f has a horizontal tangent but no local extremum, then f' touches zero without changing sign.

Common mistakes

Confusing a large function value with a large derivative value.

Thinking every horizontal tangent gives an extremum.

5.9 Connecting a Function, Its First Derivative, and Its Second Derivative

Concept

This section combines everything:

$$f' \text{ tells increasing/decreasing}$$
$$f'' \text{ tells concavity}$$
$$f' = 0 \text{ may indicate extrema}$$
$$f'' = 0 \text{ may indicate inflection points}$$

Worked Example

If f'(x) has a local maximum at x=c, then f''(c)=0 is a likely candidate because f'' is the derivative of f'.

If f'(x)>0 and f''(x)<0, then f is increasing but flattening out.

Common mistakes

Reading concavity directly from f' without first interpreting whether f' is increasing or decreasing.

Assuming f''>0 means the function itself is increasing.

5.10 Introduction to Optimization Problems

Concept

Optimization means maximizing or minimizing a quantity under a constraint.

The essential strategy is always the same: write the quantity to optimize as a function of one variable, find critical points, and test them.

Standard workflow

1. Define variables clearly.
2. Write the objective function.
3. Use the constraint to rewrite the objective in one variable.
4. Find critical points.
5. Justify maximum or minimum with a test or endpoint check.

Worked Example

A rectangle with perimeter 20 has side lengths x and y. Then

$$2x + 2y = 20 \Rightarrow y = 10 - x$$
$$A = xy = x(10 - x) = 10x - x^2$$
$$A'(x) = 10 - 2x = 0 \Rightarrow x = 5$$

So the rectangle of maximum area is a square, 5\times 5.

Common mistakes

Optimizing the wrong quantity.

Stopping after finding a critical point without checking whether it is a maximum or minimum.

5.11 Solving Optimization Problems

Concept

On AP problems, optimization often appears in geometry or applied settings such as fencing, boxes, cost, surface area, or shortest distance.

Hard Example

A farmer has 200 meters of fencing to build a rectangular pen along a river, so only three sides need fencing. Let the two equal widths be x and the side parallel to the river be y. Then

$$2x + y = 200 \Rightarrow y = 200 - 2x$$
$$A = xy = x(200 - 2x) = 200x - 2x^2$$
$$A'(x) = 200 - 4x = 0 \Rightarrow x = 50$$
$$y = 200 - 2(50) = 100$$

So the maximum area occurs at dimensions 50\times 100.

Common mistakes

Failing to rewrite everything in one variable.

Ignoring domain restrictions such as positive lengths.

Not translating the final answer back into the original context.

5.12 Exploring Behaviors of Implicit Relations

Concept

When a relation is given implicitly, such as x^2+xy+y^2=7, you use implicit differentiation to analyze tangent lines, extrema, and other local behavior.

Key formulas

$$\frac{d}{dx}\left(y^n\right) = ny^{n-1}\frac{dy}{dx}$$
$$\frac{d}{dx}(xy) = x\frac{dy}{dx} + y$$

Horizontal and vertical tangents

For an implicitly defined curve, solve for \frac{dy}{dx}.

A horizontal tangent occurs when the numerator is zero and the denominator is nonzero.

A vertical tangent occurs when the denominator is zero and the numerator is nonzero.

Worked Example

For

$$x^2 + xy + y^2 = 7$$

differentiate:

$$2x + x\frac{dy}{dx} + y + 2y\frac{dy}{dx} = 0$$

$$(x + 2y)\frac{dy}{dx} = -(2x + y)$$

$$\frac{dy}{dx} = -\frac{2x + y}{x + 2y}$$

Horizontal tangents occur when

$$2x + y = 0, \quad x + 2y \neq 0$$

Vertical tangents occur when

$$x + 2y = 0, \quad 2x + y \neq 0$$

Common mistakes

Forgetting to multiply by \frac{dy}{dx} when differentiating a y-term.

Finding points from the numerator or denominator and forgetting to check whether the point actually lies on the curve.

Unit 5 Formula Sheet

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$f'(c) = 0 \quad \text{or} \quad f'(c) \text{ DNE}$$

$$f'(x) > 0 \Rightarrow f \text{ increasing}$$

$$f'(x) < 0 \Rightarrow f \text{ decreasing}$$

$$f' : + \rightarrow - \Rightarrow \text{local max}$$

$$f' : - \rightarrow + \Rightarrow \text{local min}$$

$$f''(x) > 0 \Rightarrow f \text{ concave up}$$

$$f''(x) < 0 \Rightarrow f \text{ concave down}$$

$$f''(c) > 0 \Rightarrow \text{local min}$$

$$f''(c) < 0 \Rightarrow \text{local max}$$

$$\frac{dy}{dx} = 0 \Rightarrow \text{horizontal tangent}$$

$$\frac{dy}{dx} \text{ undefined} \Rightarrow \text{vertical tangent}$$

Final advice: separate the jobs. First locate candidates. Second decide increase/decrease or concavity from a sign chart. Third classify extrema or inflection behavior with the correct theorem or derivative test. Most errors happen when students skip one of these stages.